

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) All questions are compulsory.
(2) Make suitable assumptions wherever necessary and state the assumptions made.
(3) Answers to the same question must be written together.
(4) Numbers to the right indicate marks.
(5) Draw neat labeled diagrams wherever necessary.
(6) Use of Non-programmable calculator is allowed.

1. **Attempt any two of the following:** 12
 - a. Explain the role of application frameworks in the context of ASP.NET Core. How do they facilitate web application development?
 - b. Provide an overview of the initial steps involved in creating your first ASP.NET Core project.
 - c. Explain the essentials of debugging ASP.NET Core projects.
 - d. Why is unit testing important in the development of ASP.NET Core applications?
2. **Attempt any two of the following:** 12
 - a. Describe the concept of pagination in a web application. Why is it important? How can you implement in a Blazor server?
 - b. Discuss the key components and data structures involved in implementing a shopping cart.
 - c. What are navigation controls in a context of web application? Why are they necessary?
 - d. Explain the role of managing orders in an e-commerce application administration part.
3. **Attempt any two of the following:** 12
 - a. Explain the concept of conventional administration features in the context of security in ASP.NET Core.
 - b. Explain the key components of an ASP.NET Core project including Startup.cs and Program.cs
 - c. Explain the concept of service location and tight coupling in the context of routing in ASP.NET Core.
 - d. Describe how exceptions and errors can be handled in ASP.NET Core.
4. **Attempt any two of the following:** 12
 - a. How is Entity Framework Core used to work with data in a .NET application?
 - b. Describe the process of adding a data model to your project. Why is this important?
 - c. How can tools like Swagger and Postman assist in working with web services?
 - d. What is the significance of RESTful web services in modern web development?

5. Attempt any two of the following: 12
- a. What is the purpose of views in web development, and how do they relate to controllers?
 - b. How does TempData differ from ViewBag in terms of data storage and retrieval?
 - c. What are view components? How do they differ from regular controllers and views?
 - d. Discuss the advanced features available for tag helpers.
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